4. Installation Energy

Energy Resilience Policies and Guidance

In managing an energy resilient enterprise, the DoD must be prepared for and be able to recover from utility disruptions that impact mission assurance on its installations. DoD relies on commercial power to conduct missions from its installations, and these commercial power supplies can be threatened by physical attacks, cyberattacks, climate change, and other environmental factors. Climate change is expected to increase risks of wildfires, winter storms, hurricanes, storm surges, rain and heat hazards, and other extreme weather events, which are threat multipliers and degrade installation functions and infrastructure. The DoD recognizes that such events could result in power outages affecting critical DoD missions and continues to work to understand and address the vulnerabilities that could lead to such events and impact mission readiness.

As defined by Title 10 U.S.C. 101(e)(6), energy resilience is

"the ability to avoid, prepare for, minimize, adapt to, and recover from anticipated and unanticipated energy disruptions in order to ensure energy disruptions in order to ensure energy availability and reliability sufficient to provide for mission assurance and readiness, including mission essential operations related to readiness, and to execute or rapidly reestablish mission essential requirements."

Energy resilience can be enhanced in a variety of ways, including the use of technologies, policies, readiness exercises, and energy resilience management best-practices. Mission capability concerns override preferences for specific technology implementation goals. An important aspect of energy resilience is to establish an iterative planning and implementation cycle in which mission owners conduct risk analyses and specify requirements, infrastructure stakeholders solve for the specified requirements, and the process repeats itself as needed to meet changing mission parameters.

DoD's primary policy for energy strategy and guidance is DoD Instruction (DoDI) 4170.11, Installation Energy Management which provides a framework for reducing energy consumption and investing in energy reduction measures, with the goal of providing resilient energy to support the warfighter. A significant update to DoDI 4170.11 began in 2022 and is expected to be completed in 2024, to incorporate energy resilience strategies and climate resilience goals for the DoD.

DoD publishes policies, guidance, and the status of its energy resilience program on this website: OUSD(A&S) - ASD(EI&E) (osd.mil).

Installation Energy Plans (IEPs): Title 10 U.S.C. § 2911

Title 10 U.S.C. § 2911(b)(1) and (2) requires the DoD and Components to develop comprehensive master plans for the achievement of DoD energy goals. The DoD continues to ensure the use of structured, rigorous evaluations of critical energy requirements, risks, and mitigations by completing Installation Energy Plans (IEPs). IEPs drive a more integrated and systematic approach to energy and water management through informed energy and water resilience planning and support a more holistic energy project implementation with input from all stakeholders. Installations are responsible for conducting gap analysis and developing appropriate options for risk mitigation, while developing energy projects that optimize energy efficiency, reliability, and resiliency. In FY 2023, the DoD continued completion of IEPs across the enterprise, focusing installation resilience requirements on the most advantageous

technology and funding execution paths. To date, the DoD has completed IEPs at 73 percent of installations.

Black Start Exercises (BSEs)

BSEs are a mission-assurance focused exercise mandated by 10 U.S.C. § 2920 in which Military Departments must carry out no fewer than five (5) planned BSEs per year through FY 2027. As a planned, risk-mitigated exercise, BSEs emulate a loss of utility power by disconnecting an installation from off-installation energy sources. These exercises identify gaps in critical mission requirements and in conjunction with IEPs assist installation personnel in identifying and executing energy resilience solutions to support critical missions.

While each exercise has been different, general findings and resilience gaps identified from prior BSEs include:

- Backup generators alone can be insufficient to provide energy resilience with challenges that include being severely under-loaded, electrical misconfiguration, and improper prioritization of critical loads;
- Lack of resilient information technology (IT) architecture for cyber-security; and
- Unknown energy and control system interdependencies that lead to unanticipated failures.

In FY 2023, the DoD completed 23 BSEs across the Services identifying energy vulnerabilities and solutions to improve installation energy resilience.

Table 2. BSEs for Determining Energy Vulnerabilities

Service	FY 2023 BSE SNAPSHOT
ARMY	Six (6) BSEs in FY 2023: Fort Campbell, Fort Carson, Fort Drum, and U.S. Army Garrison Bavaria (which included BSEs conducted at the main campus and geographically separated sub-Installations - Grafenwoehr Training Area and Hohenfels Training Area).
NAVY	Ten (10) BSEs in FY 2023: Naval Air Station (NAS) Meridian, Naval Support Activity (NSA) Naples, NAS Jacksonville, NAS Key West, NAS Joint Reserve Base New Orleans, Naval Submarine Base (NSB) Kings Bay, NSA Panama City, NSA Hampton Roads, Naval Station Everett, and NSA Mid-South.
USMC	Two (2) BSEs in FY 2023: Marine Corps Air Station (MCAS) Yuma and Marine Corps Air Ground Combat Center (MCAGCC).
DAF	Five (5) BSEs in FY 2023: Beale Air Force Base, Robins Air Force Base, Creech Air Force Base, Otis Air National Guard, and Cape Cod Space Force Station.

Critical Energy Capacity and Availability: Title 10 U.S.C. § 2920

Critical missions, as defined in title 10 U.S.C. § 2920, refer to those missions of an installation, including mission-essential operations, that are critical to successful performance of the strategic national defense mission. Energy systems used to support critical missions are developed and managed at the installation level by each of the Components and expected to meet the requirements outlined in title 10 U.S.C. § 2920. Figure 5 below provides details on critical energy system capacity and progress toward achieving a minimum level of 99.9 percent energy system reliability for critical missions by 2030. The average system reliability, defined as the percentage of time the energy system is functional, is approximately 99.6 percent for the entire DoD.

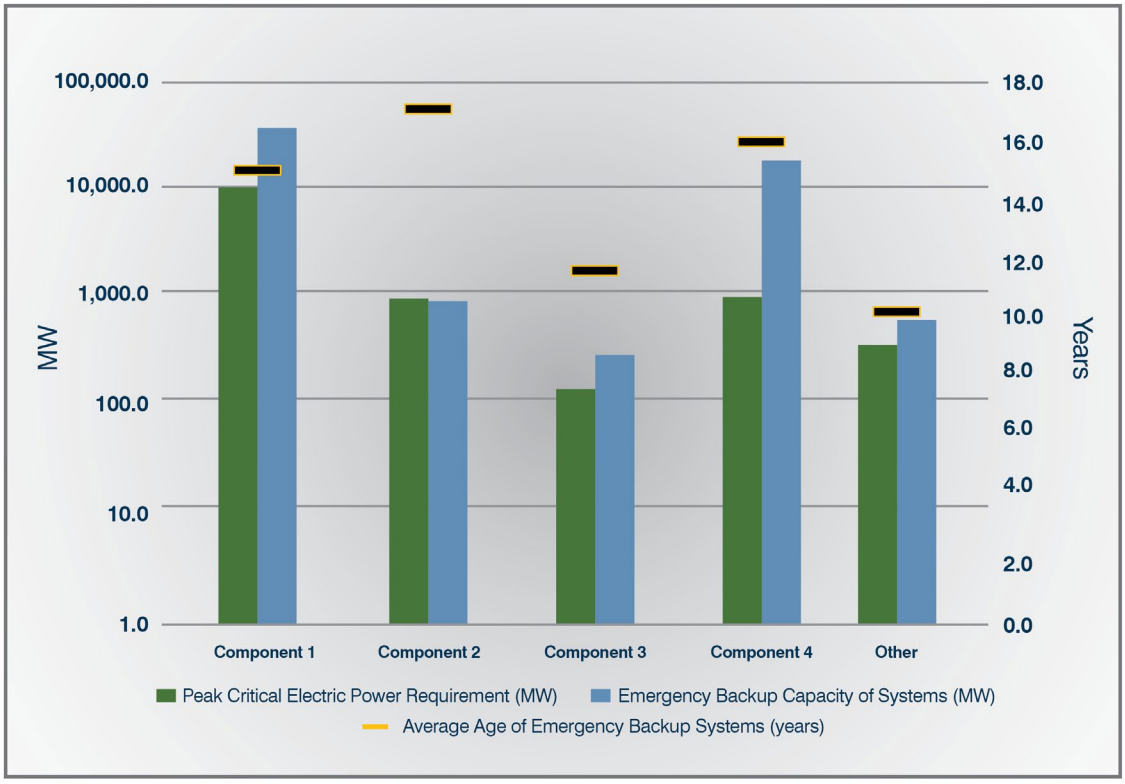


Figure 5: Critical Energy Requirements and Emergency Backup System Capacities and Average Age by DoD Component

Figure 6 below provides details on average system downtimes and average availability over the course of the year of these critical energy systems for the DoD Components.

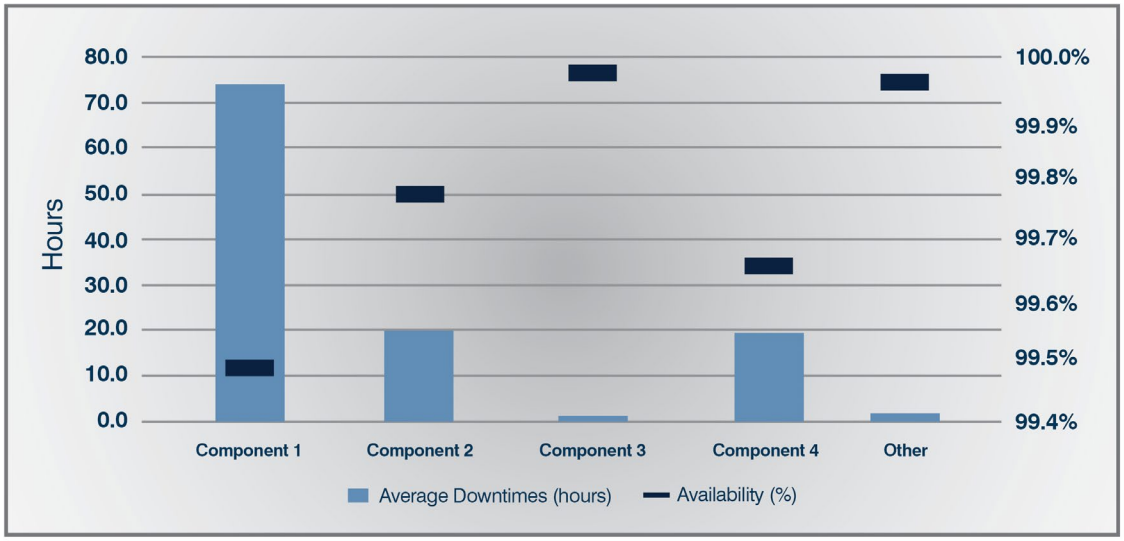


Figure 6: Average downtimes for critical energy system and average energy system availability by DoD Component

Utility Outages on Privatized Systems: Title 10 U.S.C. § 2688(g)(4)

To meet statutory requirements, DoD began collecting outage data to measure progress on privatized systems in FY19. Table 3 below provides details of FY 2023 outage data on privatized systems to help meet this requirement. The average cost of outages has increased over the past three years due to extreme weather events that affected energy systems, including Winter Storm Uri in FY 2021, heat waves across the U.S. in summer of FY 2022, wildfires on Maui Island and Typhoon Mawar in Guam in FY 2023, and other external factors affecting the cost of energy.

Table 3. Privatized System Outage Summary

PRIVATIZED SYSTEM OUTAGE SUMMARY					
	FY 2019	FY 2020	FY 2021	FY 2022	FY 2023
Percent of outages on privatized systems	19%	21%	15%	15%	19%
Percent of privatized system outages (8+ hours)	13%	17%	14%	16%	16%
NOMINAL COSTS ($)					
Total cost of outages on privatized systems	$ 404,071	$ 190,725	$76,507,606	$ 2,975,694	$ 8,620,333
Average cost per day of outages	$ 981	$ 540	$ 62,755	$ 2,190	$ 8,449

The Energy Resilience and Conservation Investment Program (ERCIP)

ERCIP, which is authorized by title 10 U.S.C. § 2914, is a critical element of the DoD's strategy to improve the energy resilience, energy security, and energy conservation of its fixed installations. ERCIP is the principal mechanism applied to execute DoD high-priority, high-value projects that improve energy resilience and mission readiness. The FY 2021 NDAA implemented significant 10 U.S.C. § 2914 modifications, transitioning ERCIP to a project/line-item authorized Military Construction (MILCON) program. ERCIP improves readiness by focusing on the closure of critical energy gaps at priority locations, increasingly through the installation of flexible and expandable microgrids, deployment of energy storage technologies, the leveraging of existing renewable energy generation resources, and application of cyber-security improvements. As ERCIP projects employ increasingly complex technologies, the cost to design and construct ERCIP projects also increases. In FY 2023, Congress added $100M to ERCIP, increasing the FY 2023 appropriations to just over $653M. In FY 2024, just over $634M was requested and appropriated for ERCIP.

Table 4. Energy Resilience Project Snapshots

Service	FY 2023 ERCIP SNAPSHOT
ARMY	In FY 2023, the Army received $119M for five power generation and microgrid projects. Additionally, the Army received $36.8M in cost to complete funds for four FY 2022 projects and began the detailed planning and design work for 14 projects.
NAVY	The Navy received $132.72M for eight projects at seven locations. This included critical electrical distribution infrastructure upgrades at Naval Base (NAVBASE) Guam and Joint Base Pearl Harbor-Hickam, Hawaii. The Navy also received approximately $32M in cost to complete funds for two projects in Guam and began design efforts for seven projects.
USMC	In FY 2023, the USMC received $25.5M for Marine Corps Mountain Warfare Training Center Bridgeport, California. $608K in cost to complete funds were provided for MCAS Miramar to complete construction on the landfill gas (LFG) Power Meter Station. An additional three projects received funding to begin design efforts.
DAF	The Department of Air Force received $15.7M for two projects at Patrick Space Force Base (SFB), Florida. The DAF received an additional $30M in cost to complete funds for two FY 2022 projects and began design efforts for nine projects at eight locations.

Installation Energy Alternative Funding

The DoD seeks to maximize the use of authorities to pursue third-party financing of installation efficiency, renewable, other carbon pollution-free (CFE) energy generation, and resilience projects. These authorities are Utility Energy Service Contracts (UESCs) and Energy Savings Performance Contracts (ESPCs) (title 10 U.S.C. § 2913), power-purchase agreements (PPAs) (title 10 U.S.C. § 2922a), and outgrants (title 10 U.S.C. § 2667 and 10 U.S.C. § 2668)(7) (see Appendix H).

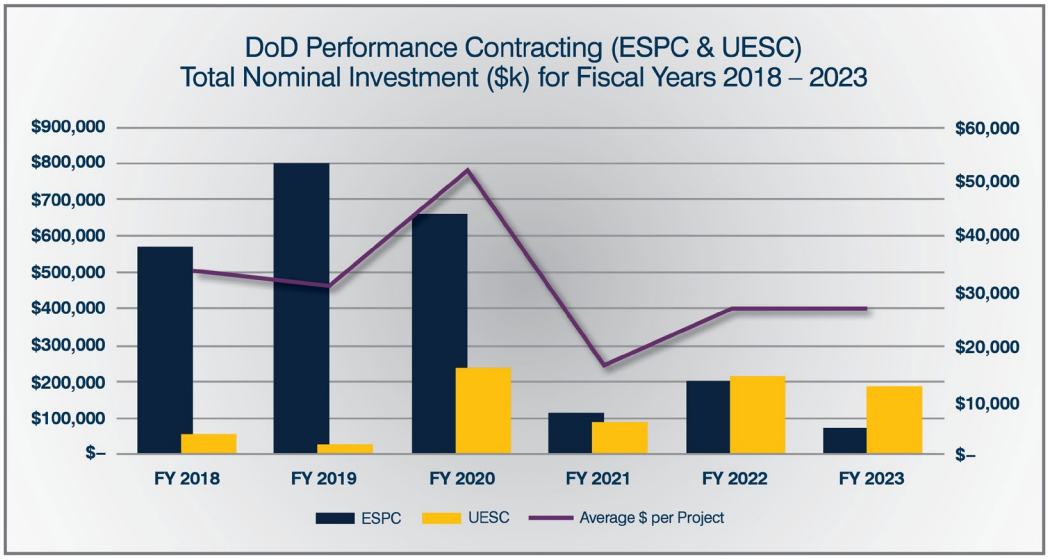


Figure 7: DoD Performance Contracting (ESPC and UESC) Total Nominal Investment Value ($k) for Fiscal Years 2018 – 2023

The DoD has the largest performance contracting portfolio in the Federal Government (over $5B). As enabled by the Energy Act of 2020, Energy Independence and Security Act (EISA) 2007, and Energy Policy Act (EPAct) 2005, DoD continues to use performance contracts to bolster resilience and reduce energy and water intensity. By statute, all such projects must pay for themselves through energy savings. The average size of DoD contracts is comparatively large in comparison to other Federal agencies ($32M

average per project for FY 2018 – FY 2023, Figure 7 above). From FY 2018 – FY 2023 the Department awarded just over $3.2 billion through 101 performance contracts. In FY 2023, the DoD awarded three ESPCs with a total investment value of $74.4M, and seven UESCs with a total investment value of $192.6M across all the Components. In FY 2023, ODASD(ER&O) began evaluating strategies to align third-party financing with building level efforts (i.e., Deep Energy Retrofit (DER), Electrification, Audits, etc.), and coordinating with the Military Departments to increase the third-party financing pipeline.

Table 5. Alternative Financing Project Snapshots

Service	FY 2023 ALTERNATIVE FINANCING SNAPSHOT
ARMY	In FY 2023, the Army awarded $20.4M across two ESPC projects and $135.8M across four UESC projects. These projects will reduce greenhouse gas (GHG) emissions by 63,386 tons carbon dioxide (CO_2) equivalent, include 3.5 Megawatt (one million watts) (MW) of CFE generation, and 7.5 megawatt hours (MWh) of battery energy storage. The Army partnered with the Department of Energy (DOE) to provide Army-specific ESPC and UESC training to new project staff to increase execution capacity.
NAVY	The Navy was granted approval by OSD to utilize 10 U.S.C. 2913(c) to allow Naval Base San Diego to accept incentives in the form of goods and services. On 20 September 2022, Navy Region Southwest entered into an Intergovernmental Support Agreement (IGSA) with the Port of San Diego to register and generate Low Carbon Fuel Standard (LCFS) credits based on the Navy vessels' use of shore power in port that meets the requirements of the LCFS regulation.
USMC	In FY 2023, MCAS Beaufort awarded a $23.7 million UESC to Dominion Energy for electrical distribution and lighting improvements. The UESC will contribute to the resilience of MCAS Beaufort by providing funding to upgrade and reconfigure over 15 miles of distribution lines from overhead to underground and complete lighting upgrades. The contract leverages $20.2M ERCIP cost avoidance funding leaving a capital cost of $3.5M, before financing. The contract will be financed over 22.5 years, including a 29-month construction period.
DAF	Joint Base San Antonio reported decreased consumption by 18.6 percent from the previous year. This decrease is a direct result of the fully commissioned energy conservation measures from the last ESPC contract.

Installation Renewable Energy Goals

The DoD currently works to achieve two annual Federally mandated renewable energy goals: Section 203 of the EPAct 2005 (title 42 U.S.C. § 15852(a)) and title 10 U.S.C. § 2911(g). Section 319(c)(2) of the NDAA for FY 2022, (Public Law 117-81) requires that the DoD provide an annual briefing on the progress made toward meeting the goal described in title 10 U.S.C. § 2911(g)(1)(A), with respect to FY 2025. This section and the following section (Renewable Energy Project Deployment and Carbon Pollution-Free Electricity (CFE)) satisfy that briefing request.

The goal outlined in title 10 U.S.C. § 2911(g) is specific to DoD, and progress toward the goal cannot include the purchase of unbundled renewable energy credits (RECs). DoD progress toward both installation renewable energy goals for FY 2023 is shown in Table 6.

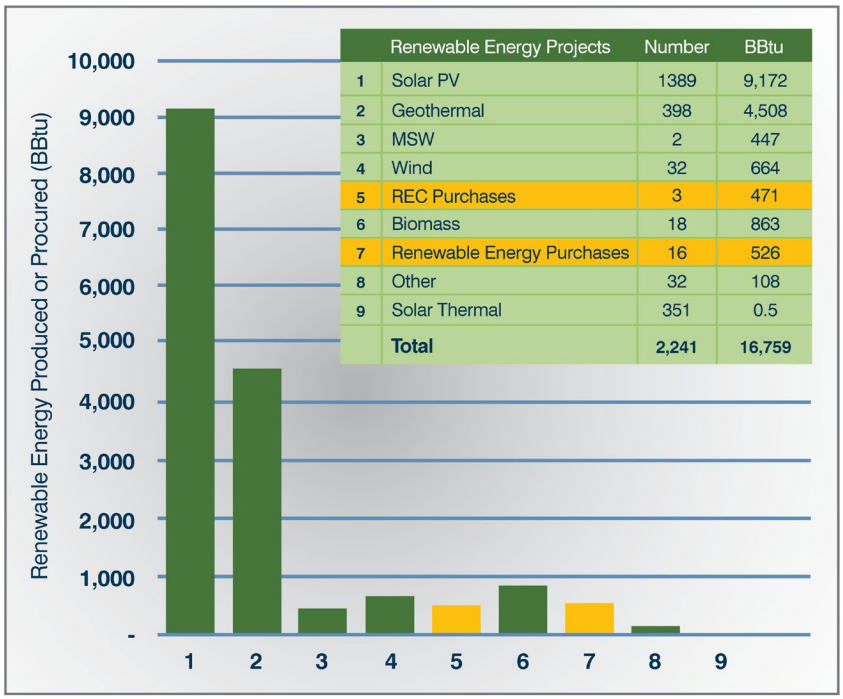
Table 6. FY 2023 Progress Toward Installation Energy Goals

GOALS AND OBJECTIVES	METRIC	COMPONENT	FY 2023	GOAL
Consume More Electric Energy from Renewable Sources Title 42 U.S.C. § 15852(a) – EPAct 2005	Total renewable electricity consumption as a percentage of total facility electricity consumption.	DoD	7.5%	7.5%
		ARMY	10.6%	
		NAVY	5.9%	
		USMC	11.8%	
		USAF	6.6%	
Produce or Procure More Energy from Renewable Sources Title 10 U.S.C. § 2911(g)	Total renewable energy (electric and non-electric) produced or procured as a percentage of total facility energy consumption.	DoD	17.9%	25% by 2025
		ARMY	12.9%	
		NAVY	33.6%	
		USMC	9.4%	
		USAF	17.7%	

Renewable Energy Project Deployment and Carbon Pollution-Free Electricity (CFE)

DoD pursues renewable energy and other CFE to support and advance installation energy and climate resilience, while also complying with legal requirements to increase its renewable energy supply. DoD continues to invest in an array of cost-effective renewable and distributed energy solutions such as solar photovoltaic (PV), wind, and geothermal technologies, as well as deploying on-site energy storage solutions. The DoD focuses renewable energy investments using IEPs and Service installation-level assessments to identify locations and facilities where the greatest benefit in energy resilience improvement and greenhouse gas emissions reduction can be achieved. Figure 8 below demonstrates DoD's commitment and progress towards renewable energy goals, DoD renewable energy generation increased by roughly 25 percent in FY 2023 (16,759 BBtu) as compared to FY 2022 (13,401 BBtu)[13].

[13] See FY 2022 AEPRR Report, page 30 (Figure 8). Available here: https://www.acq.osd.mil/eie/Downloads/IE/FY22%20AEPRR%20Report.pdf



	Renewable Energy Projects	Number	BBtu
1	Solar PV	1389	9,172
2	Geothermal	398	4,508
3	MSW	2	447
4	Wind	32	664
5	REC Purchases	3	471
6	Biomass	18	863
7	Renewable Energy Purchases	16	526
8	Other	32	108
9	Solar Thermal	351	0.5
	Total	**2,241**	**16,759**

Figure 8. FY 2023 Total Renewable Energy Projects and Energy Produced or Procured in BBtu by Type

As detailed in the *Department of Defense Carbon Pollution-Free Electricity Strategic Plan for FY 2024*, the Department obtained 46 percent of its electricity from CFE sources in FY 2023. DoD aims to transition to 100 percent CFE on an annual basis by FY 2030, with at least 50 percent of demand matched to CFE regional supply on an hourly basis in accordance with EO 14057.[14] To achieve this goal, in FY 2022 the DoD began an evaluation of changes to organizational structures and acquisition methods. In FY 2023, DoD has coordinated closely with other government agencies, especially the General Services Administration (GSA) and the DOE, in a "whole-of-government" approach that leverages the purchasing power of the U.S. Government, as well as combining Service-level contracts to increase purchasing power and encourage utilities to source additional CFE.

The DoD uses various authorities to increase the supply of distributed (on-site) and renewable energy sources on its installations. DoD uses both appropriated funds and non-Governmental ("third-party" or "alternative") financing to pursue renewable energy projects. Typically, DoD contracts with non-Federal entities to enable the development of large-scale renewable (or other distributed) energy projects, using third-party financing to fund the initial project costs. DoD also relies on Congressional appropriations to fund cost-effective, small-scale distributed generation projects that do not require third-party financing.

In FY 2023, the DoD made progress on multiple fronts with respect to CFE:

- Launched Pathfinders and other engagements covering ~40 percent of CFE need (up to 183 MW in Carolinas, ~75 MW in Florida, and ~36 MW in Arkansas);
- Continued to advance onsite CFE development (~300 MW in development); and
- Developed a CFE Policy to define roles and responsibilities for CFE (to be issued FY 2024).

[14] E.O 14057: https://www.whitehouse.gov/briefing-room/presidential-actions/2021/12/08/executive-order-on-catalyzing-clean-energy-industries-and-jobs-through-federal-sustainability/

Table 7. Renewable Energy Projects

Service	FY 2023 Renewable Energy Project SNAPSHOT
ARMY	In FY 2023, the Army added eight new renewable energy projects to its portfolio for a combined total of 1.3 MW of rated capacity. Additionally, a new 28 MW solar microgrid project, completed in FY 2023 at the Joint Forces Training Base in Los Alamitos, will begin generating power in FY 2024.
NAVY	The Navy executed two Enhanced Use Leases (EULs) to develop renewable energy power generation projects on Joint Base Pearl Harbor-Hickam (JBPHH). These projects will provide 250+ MW of renewable energy to the local utility, Hawaiian Electric Company (HECO), in support of island grid reliability, energy gap mitigation, and mission continuance in the event of grid outage.
USMC	MCAS Beaufort was competitively selected for a DOE Assisting Federal Facilities with Energy Conservation Technologies (AFFECT) grant for $7.8M. The AFFECT funding supports the installation of solar PV carport over the parking lot adjacent to the gymnasium and dining facility.
DAF	The major element to DAF renewable energy performance improvement was the installation of a 460 MW solar EUL located on Edwards Air Force Base (AFB) which all the power was sold into the state grid and a percentage of the rent revenues may be used to fund energy-related projects at Edwards AFB.

Availability and Use of Energy Cost Savings: Title 10 U.S.C. § 2912

Title 10 U.S.C. § 2912 allows the DoD Components to reinvest energy cost savings. In FY 2023, the Components awarded over 150 projects totaling $53.1M in reinvested energy cost savings. Table 8 includes highlighted FY 2023 projects and activities from the Services.

Table 8. Energy Cost Savings Projects

Service	FY 2023 2912 Project SNAPSHOT
ARMY	In FY 2023, the Army captured more than $49 million of validated installation energy cost savings. Over $44 million of those savings were reinvested into qualified projects before the end of FY 2023.
NAVY	In FY 2023, the Navy continued its implementation of the 10 U.S.C. § 2912 Pilot Program through the Navy Energy Headquarters Program Director that was launched in FY 2021. This program incentivizes and rewards installations for implementing and leading energy consumption reduction efforts by re-investing savings back into all Navy installations. Over the last three years, the Navy's 2912 program has earned $100+M and funded a total of 290 energy projects, advancing the Navy's energy goals and objectives.
USMC	In FY 2023, the USMC recovered $1.2 million. The amount recovered funds was limited by the Assistant Secretary of the Navy (Financial Management and Comptroller).
DAF	The Air Force did not harvest energy cost savings or award any projects in FY 2023, but they continued to award and execute projects from the existing "no-year" funds captured before the statute was amended in the FY 2022 NDAA[15].

The authorized uses of harvested energy cost savings for reinvestment are defined in 10 U.S.C. § 2912(b)(1), § 2912(b)(2), and § 2912(c) and listed in Table 9 below.

[15] Section 315 of the FY 2022 NDAA changed the language of 10 U.S.C. § 2912 in subsection (a) by striking "until expended" and inserting "for that fiscal year and the succeeding fiscal year." Funds previously transferred remain available for obligation for section 10 U.S.C. § 2912 purposes and retain their "no-year" period of availability.

Table 9. Authorized Uses for Energy Cost Savings Reinvestment

Installation Energy		Operational Energy
Enterprise Funds - 10 U.S.C. § 2912(b)(1)	Installation Funds - 10 U.S.C. § 2912(b)(2)	Enterprise Funds - 10 U.S.C. § 2912(c)
Installation energy resilience	Improvements to existing military family housing units	Operational energy resilience
Mission assurance	Unspecified minor construction project that will enhance the quality of life of personnel	Efficiencies
Weather damage repair and prevention		Mission assurance
Energy conservation	Morale, welfare, or recreation facility or service	Energy conservation
Energy security measures		Energy security.

In accordance with 10 U.S.C. § 2912(e)(3), DoD Components are also required to provide a detailed description of the funds are transferred between appropriations/authorizations (10 U.S.C. § 2912(e)(1)) and the purpose for which such amounts have been used (as shown in Table 10).

Table 10. FY 2023 Transfers Between Appropriations/Authorizations (10 U.S.C. 2912(e)(1))

Service	Funds Transferred	Transferred From	Transferred To	Funds Use Description
ARMY	$857,000	206520D22	204020D22	All-terrain electric Mission Module trailers
	$13,666,000	202020D22	204020D22	All-electric Infantry Squad Vehicles and an H2-fuel cell powered mobile charging station
	$426,000	202020D22	204020D22	All-terrain electric Mission Module trailers
USMC	$600,000	1711062222D	1718042223D	Update the main controllers at Marine Corps Air Ground Combat Center (MCAGCC) Energy Program in all the buildings required to get these systems back on the network and operational. Support security and energy savings.

Facility-Related Control Systems

In FY 2023, Deputy ASD(EI&E) provided funding for the planning and site visits needed to conduct Cybersecurity Resilience Readiness Exercises (CRREs) at four mission critical installations in FY 2024. These CRREs build on a successful live cyber resilience readiness exercise, under sponsorship of ODASD(ER&O), that was performed on an active DoD installation over a 12-hour period in FY 2021, in conjunction with a BSE. The privatized utility power was disconnected via cyber means and demonstrated two different cyber effects on both mission-owned and real-property control systems. The findings from the planning, execution, analysis, and reporting from this live-fire cyber exercise were impactful and actionable for the installation, critical mission partner, and military department that hosted the exercise.

The success of this initial exercise demonstrated the broad range of control systems on an installation that can impact mission assurance; the variability in understanding, coordination, and ownership of the relevant systems; and the limited guidance, tools and resources that are available for installations to identify the unique cyber-physical vulnerabilities of their control systems. Analogous to BSEs, CRREs will be conducted with each Service to inform the development of Control Systems Cybersecurity policy, Unified Facilities Criteria and a CRRE playbook outlining execution of the exercises as a mandatory effort whose outputs inform mission impact, risk mitigation and investment.



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Appendix A - List of Acronyms

Acronym	Definition
AEMRR	Annual Energy Management and Resilience Report
AEPRR	Annual Energy Performance, Resilience, and Readiness Report
AETP	Advanced Engine Transition Program
AF/A4	HQ USAF Deputy Chief of Staff for Logistics, Engineering, and Force Protection
AFB	Air Force Base
AFFECT	Assisting Federal Facilities with Energy Conservation Technologies
AFSOC	Air Force Special Operations Command
AMI	Advanced meter infrastructure
AMMPS	Advanced Medium Mobile Power Source
AOR	Area of responsibility
APS	Advanced power strips
ARC	Army Reserve Center
ARCENT	Army Central Command
ARNG	Army National Guard
ASA(IE&E)	Assistant Secretary of the Army for Installations, Energy and Environment
ASD(EI&E)	Assistant Secretary of Defense for Energy, Installations and Environment
ASN(EI&E)	Assistant Secretary of the Navy for Energy, Installations and Environment
ATO	Air Tasking Order
ATS	Automatic Transfer Switch
BBtu	Billion British thermal units
BESS	Battery Energy Storage System
BOS	Base operations support
BSE	Black Start Exercise
Btu	British thermal unit
CAAFI	Commercial Aviation Alternative Fuels Initiative
CAP	Climate Action Plan
CO_2	Carbon dioxide
CCMD	Combat Commands
CFE	Carbon pollution-free electricity
CAP	Climate Action Plan
CLOE WG	Contested Logistics Operational Energy Working Group

Acronym	Definition
CNIC	Commander, Navy Installations Command
CNO	Office of the Chief of Naval Operations
COOP	Continuity of Operations
CRRE	Control System Resilience Readiness Exercise
CSG	Executive Agent Component Steering Group
DAF	Department of the Air Force
DAFI	Department of the Air Force Instruction
DASA(E&S)	Deputy Assistant Secretary of the Army for Energy and Sustainability
DASN(E)	Deputy Assistant Secretary of the Navy for Energy
DC2DL	Determining Climate Challenges for Defense Logistics
DCA	Defense Critical Asset
DC I&L	USMC Deputy Commandant for Installations and Logistics
DCMA	Defense Contract Management Agency
DCS	Deputy Chief of Staff
DeCA	Defense Commissary Agency
DER	Deep energy retrofit
DFARS	Defense Federal Acquisition Regulation Supplement
DFAS	Defense Finance and Accounting Service
DIA	Defense Intelligence Agency
DLA	Defense Logistics Agency
DMAG	Deputy's Management Action Group
DMO	Distributed Maritime Operations
DoD	Department of Defense
DoDD	DoD Directive
DoDI	DoD Instruction
DOE	Department of Energy
DoN	Department of the Navy
EABO	Expeditionary Advanced Base Operations
EISA 2007	Energy Independence and Security Act 2007
eKPP	Energy Key Performance Parameter
eLRV	Electric light reconnaissance vehicle (eLRV)
Energy HPD	Energy Headquarters Program Director
EO	Executive Order

Acronym	Definition
EPAct 2005	Energy Policy Act of 2005
ERCIP	Energy Resilience Conservation and Investment Program
ERP	Energy Readiness Program
ESA	Energy Supportability Analysis
ESCO	Energy service contractor
ESPC	Energy Savings Performance Contract
ESTCP	Environmental Security Technology Certification Program
EUL	Enhanced use lease
F2EP	Office of Future Force Energy and Power
FCBs	Functional Capability Boards
FEB	Facility Energy Board
FEC	Navy Facility Engineering Command
FEG	Facility Energy Group
FEMP	Federal Energy Management Program
FEP	Facility Energy Panel
FRCS	Facility-related control systems
FEOC	Facility Energy Operations Center
FY	Fiscal year
FYDP	Future Years Defense Program
GBFMD	Global Bulk Fuel Management and Delivery
GENISYS	Navy Global Energy Information System
GSA	General Services Administration
GVSC	Army Ground Vehicle Systems Center
HAF	Headquarters Air Force
HASC	House Armed Services Committee
HECO	Hawaiian Electric Company
HMMWV	High Mobility Multipurpose Wheeled Vehicle
HQ	Headquarters
HVAC	Heating, ventilation, and air conditioning
IAPR	Integrated Acquisition Portfolio Reviews (IAPRs)
ICAO	International Civil Aviation Organization
IE	Installation energy
IEP	Installation Energy Plan

Acronym	Definition
IGSA	Intergovernmental Services Agreement
IMM	Integrated Material Management
IMM	Integrated Materiel Manager
IoT	Internet of things
IRP	Installation Resilience Plan
IT	Information technology
JB	Joint Base
JBPHH	Joint Base Pearl Harbor-Hickam
JLENT	Joint Logistics Enterprise
JLTV	Joint Light Tactical Vehicles
JPEC	Joint Petroleum Executive Council
JPSG	Joint Petroleum Steering Group
JSOC	Joint Special Operations Command
KW	Kilowatt, 1 thousand Watts
kWh	Kilowatt-hours
LC	Liquid chromatography
LCFS	Low Carbon Fuel Standard
LED	Light-emitting diode
LFG	Landfill gas
LMP	Liquid monopropellant
MAF	Mobility Air Forces
MCAGCC	Marine Corps Air Ground Combat Center
MCAS	Marine Corps Air Station
MCICOM	Marine Corps Installations Command
MCICOM GF	MCICOM, Facilities Directorate
MCICOM GF-PW	MCICOM, Public Works
MCTSSA	Marine Corps Tactical Systems Support Activity
MEEP	Air Force Mission Execution Excellence Program
MDA	Missile Defense Agency
MILCON	Military Construction
MMBtu	Million British thermal units
MOSAICS	More Situational Awareness for Industrial Control Systems
MOTCO	Military Ocean Terminal Concord

Acronym	Definition
MSIC	Missile and Space Intelligence Center
MW	Megawatt, 1 million watts
MWh	Megawatt-Hour, 1 million watt-hours
NAS	Naval Air Station
NAVAIR	Naval Air Systems Command
NAVBASE	Naval Base
NAVFAC	Naval Facilities Engineering Command
NAVSEA	Naval Sea Systems Command
NDAA	National Defense Authorization Act
NDS	National Defense Strategy
NGA	National Geospatial-Intelligence Agency
NGAD	Next Generation Air Dominance
NGAP	Next Generation Adaptive Propulsion Program
NH139	Joint Use Intelligence Center
NITMRE	Next Generation Information Technology for Mobility Readiness Enhancement
NREL	National Renewable Energy Laboratory
NRO	National Reconnaissance Office
NSA	National Security Agency
NSA [*base name*]	Naval Supply Activity
NSB	Naval Submarine Base
NSWC	Navy Surface Warfare Center
NTV	Non-tactical vehicle
OASD(Acquisition)	Office of the Assistant Secretary of Defense for Acquisition
OASD(EI&E)	Office of the Assistant Secretary of Defense for Energy, Installations and Environment
OCO	Overseas Contingency Organization
ODASD(ER&O)	Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization
OE	Operational energy
OECIF	Operational Energy Capability Improvement Fund
OEPF	Operational Energy Prototyping Fund
OPNAV N4I	CNO Shore Installations Division
OPT	Operational Planning Team
OSD	Office of the Secretary of Defense
OUSD(A&S)	Office of the Under Secretary of Defense for Acquisition and Sustainment

Acronym	Definition
OIUSD(R&E)	Office of the Under Secretary of Defense for Research and Engineering
PDM	Program Decision Memorandum
PLC	Programmable logic controller
PPA	Power purchase agreement
PREPA	Puerto Rico Electric Power Authority
PWRR	Petroleum War Reserve Requirement
PWRS	Preposition War Reserve Stock
PV	Photovoltaic
RASP	Navy Replenishment at Sea Planner
RASP-CASA	Refueling and Support Package to Enable Communications and Situational Awareness
RDT&E	Research, Development, Test and Evaluation
REC	Renewable Energy Credit
REGS	Renewable energy and storage
RICE	Reciprocating internal combustion engines
SAF/IE	Assistant Secretary of the Air Force (Installations, Environment and Energy)
SAF/IEE	Deputy Assistant Secretary of the Air Force Environment, Safety and Infrastructure
SASC	Senate Armed Services Committee
SATF	Synthetic Aviation Turbine Fuels
SBIR	AFWERX Small Business Innovation Research Program
SCADA	Supervisory Control and Data Acquisition
SCO	Strategic Capabilities Office
SOTF	Security Operations Training Facility
SFB	Space Force Base
TaCV-E	Tactical and combat vehicle - electrification
THAAD	Terminal High Altitude Area Defense
TTPs	Tactics, techniques, and procedures
TEARF	Test and Evaluation of Alternative Rocket Fuels
TOC	Total Ownership Cost
TWCF	Transportation Working Capital Fund
UAS	Unmanned aerial system
UAVs	Unmanned aerial vehicles
UESC	Utility Energy Service Contract
UMCS	Utility monitoring and control system

Acronym	Definition
UP	Utilities privatization
UPS	Uninterruptible power supply
U.S.	United States
USAF	U.S. Air Force
USAR	U.S. Army Reserve
USASOCOM	U.S. Army Special Operations Command
U.S.C.	United States Code
USCENTCOM	U.S. Central Command
USCG	U.S. Coast Guard
USEUCOM	U.S. European Command
USINDOPACOM	U.S. Indo-Pacific Command
USMC	U.S. Marine Corps
USSF	U.S. Space Force
USSOCOM	U.S. Special Operations Command
USTRANSCOM	U.S. Transportation Command
WHS	Washington Headquarters Services

Appendix B - Congressional Reporting Requirements

Section 2925(a) of title 10, United States Code (U.S.C.) requires the Secretary of Defense to provide "a report detailing the fulfillment during that fiscal year of the authorities and requirements under Sections 2688, 2911, 2912, 2920, and 2926 of this title, including progress on energy resilience at military installations and the use of operational energy in combat platforms and at contingency locations."

The FY 2023 Annual Energy Performance, Resilience, and Readiness (AEPRR) Report satisfies the DoD's statutory reporting requirements pursuant to the following:

- Title 42 U.S.C. § 8258, which requires Federal agencies to describe their energy management activities in a report to the Secretary of Energy;
- Title 10 U.S.C. § 2925(a), which requires DoD to submit to Congress an AEPRR describing its installation energy activities;
- Title 10 U.S.C. § 2688 (g)(4), which requires DoD to report progress in meeting energy resilience metrics for all utility conveyance contracts entered into;
- Title 10 U.S.C. § 2911(c), which requires DoD to establish and report energy performance goals for transportation systems, support systems, utilities, and infrastructure and facilities;
- Title 10 U.S.C. § 2912 (e)(3), which requires DoD to provide a detailed description of the use of funds attained from energy cost savings realized by the Department; and
- Title 10 U.S.C. § 2926 (e)(4), which requires DoD to provide proposed operational energy budgets for each of the Services as well as an operational energy strategy and updates to the strategy every 5 years.

The Compliance Matrix in Appendix C provides all reporting requirements and requests satisfied by this report, including the following additional report requests:

- Public Law 114-328, S.2943, the National Defense Authorization Act for Fiscal Year 2017, 10 U.S.C. § 2815(f).
    - Report on Military Installation Resilience Projects, requires an annual report to the congressional defense committees 90 days after the end of each fiscal year until December 31, 2025
- Public Law 117-81, S.1605, the National Defense Authorization Act for Fiscal Year 2022:
    - Energy, Water, and Waste Net-zero Requirement for Major Military Installations. - Status Report and Briefing on Progress (Section 319(c)(2))
- Senate Report 117–130, accompanying SR 7776, the National Defense Authorization Act for Fiscal Year 2023:
    - Briefing on Overseas Energy Efficiency Study
- Senate Report 116–236, accompanying SR 4049, the National Defense Authorization Act for Fiscal Year 2021:

- Report on Installation Energy - Pilot Program on Energy Management

- House Report 118–125, accompanying HR 2670, the National Defense Authorization Act for Fiscal Year 2024:
  - Report on Propane/Alternative Fuel Powered Vehicles

- Senate Report 118–58, accompanying SR 2226, the NDAA for Fiscal Year 2024:
  - Briefing on Hydrogen Fuel Cell Research for Defense Applications
  - Briefing on Benefits and Use of Electrochromic Glass in Military Bases and Installations
  - Briefing on Ground Source Heat Pumps

Appendix C - Compliance Matrix

	Subsection	Description	FY 2023 AEPRR Chapter/Appendix	Page Number
Title 10 U.S.C. § 2925	(a)	Annual Report Related to Installations Energy Performance, Resilience, and Readiness — Not later than 240 days after the end of each fiscal year, the Secretary of Defense shall submit to the congressional defense committees a report detailing the fulfillment during that fiscal year of the authorities and requirements under sections 2688, 2911, 2912, 2920, and 2926 of this title, including progress on energy resilience at military installations and the use of operational energy in combat platforms and at contingency locations.		
	(b)(1)(A)	A description of the progress made to achieve the goals of the Energy Policy Act of 2005 (Public Law 109-58), section 2911(g) of this title, and the Energy Independence and Security Act of 2007 (Public Law 110-140).	**Chapter 4, Appendix D**	**19– 29, 42 - 43**
	(b)(1)(B)	A description of the energy savings, return on investment, and enhancements to installation mission assurance realized by the fulfillment of the goals described in subparagraph (A).	**Chapter 4**	**19– 29**
	(b)(1)(C)	A description of and progress toward the energy security, resilience, and performance goals and master planning for the Department of Defense, including associated metrics pursuant to subsections (c) and (d) of section 2911 of this title and requirements under section 2688(g) of this title.	**Appendix D**	**42- 43**
	(b)(1)(D)	An evaluation of progress made by the Department in implementing the operational energy strategy of the Department, including the progress of key initiatives and technology investments related to operational energy demand and management.	**Chapter 3**	**5 – 18**
	(b)(1)(E)	Details of the amounts of any funds transferred by the Secretary of Defense pursuant to section 2912 of this title, including a detailed description of the purpose for which such amounts have been used.	**Chapter 3**	**5 – 18**
	(b)(2)	Statistical information on operational energy demands of the Department, in terms of expenditures and consumption, for the preceding five fiscal years, including information on funding made available in regular defense appropriations Acts and any supplemental appropriations Acts.	**Chapter 3**	**5 – 18**
	(b)(3)	A description of each initiative related to the operational energy strategy of the Department and a summary of funds appropriated for each initiative in the previous fiscal year and current fiscal year and requested for each initiative for the next five fiscal years.	**Chapter 3, Appendix O**	**5 – 18, 88 – 127**
	(b)(4)	Such recommendations as the Secretary considers appropriate for additional changes in organization or authority within the Department to enable further implementation of the energy strategy and such other comments and recommendations as the Secretary considers appropriate.	**Appendix F**	**47**

	Subsection	Description	FY 2023 AEPRR Chapter/Appendix	Page Number
Title 10 U.S.C. § 2911	(c)(1)	The Secretary of Defense shall submit to the Congressional defense committees the energy performance goals for the Department of Defense regarding transportation systems, support systems, utilities, and infrastructure and facilities.	Appendix D	42 – 43
	(c)(3)	The Secretary of Defense shall include the energy security and resilience goals of the Department of Defense in the installation energy report submitted under Section 2925(a) of this title for Fiscal Year 2018 and every fiscal year thereafter. In the development of energy security and resilience goals, the Department of Defense shall conform with the definitions of energy security and resilience under this title. The report shall include the amount of critical energy load, together with the level of availability and reliability by fiscal year the Department of Defense deems necessary to achieve energy security and resilience.	Chapter 4	19– 29
	(d)(1)	The Secretary of Defense shall develop a comprehensive master plan for the achievement of the energy performance goals of the Department of Defense, as set forth in laws, executive orders, and Department of Defense policies.	Appendix D	42 – 43
Title 10 U.S.C. § 2688	(g)(4)	The Secretary of Defense, in consultation with the Secretaries of the military departments, shall include in the installation energy report submitted under Section 2925(a) of this title a description of progress in meeting energy resilience metrics for all conveyance contracts entered into pursuant to this section.	Chapter 4	19– 29
Title 10 U.S.C. § 2912	(e)(3)	At the end of each fiscal year, the Secretary of Defense shall submit to Congress a report detailing any funds transferred pursuant to paragraph (1) during that fiscal year, including a detailed description of the purpose for which such amounts have been used.	Chapter 3	5 – 18
Title 10 U.S.C. § 2920	(g)(1)	If by the end of fiscal year 2029, the Secretary determines that the Department will be unable to meet the requirements under subsection (a), not later than 90 days after the end of such fiscal year, the Secretary shall submit to the Committees on Armed Services of the Senate and House of Representatives a report detailing the projected shortfall;	Chapter 4	19– 29
	(g)(2)	reasons for the projected shortfall;	Chapter 4	19– 29
	(g)(3)	any statutory, technological, or monetary impediments to achieving such requirements;	Chapter 4	19– 29
	(g)(4)	any impact to readiness or ability to meet the national defense posture; and	Chapter 4	19– 29
	(g)(5)	any other relevant information as the Secretary considers appropriate.	Chapter 4	19– 29

	Subsection	Description	FY 2023 AEPRR Chapter/Appendix	Page Number
Title 10 U.S.C. § 2926	(e)(4)	Not later than 30 days after the date on which the budget for fiscal year 2024 is submitted to Congress pursuant to section 1105 of title 31, and every five years thereafter, the Assistant Secretary shall submit to the congressional defense committees the strategy required under paragraph (1).	**Chapter 3**	**5 – 18**
Public Law 117-81, S.1605, the National Defense Authorization Act for Fiscal Year 2022	Section 319(c)(2)	During fiscal year 2022 and each succeeding fiscal year through fiscal year 2025, the Secretary shall provide to the Committees on Armed Services of the House of Representatives and Senate a briefing on the progress the Secretary has made toward meeting the goal described in section 2911(g)(1)(A) of title 10, U.S.C., with respect to fiscal year 2025.	**Chapter 4**	**Renewable Energy Goals / Project Deployment**
10 U.S.C. § 2815. FY17 NDAA	(f)	Military installation resilience projects, requires an annual report to the congressional defense committees 90 days after the end of each fiscal year until December 31, 2025	**Appendix P**	**128 - 149**
Report 117-130: Senate Armed Services Committee (SASC) FY23 Committee Report	Page 353	Overseas Energy Efficiency Study	**Appendix I**	**57 – 68**
Report 116-236: SASC FY21 Committee Report	Page 180	Installation Energy - Pilot program on Energy management	**Appendix J**	**69 – 71**
Report 118-125: HASC FY24 Committee Report	Page 100	Propane/Alternative Fuel Powered Vehicles	**Appendix K**	**72 – 74**
Report 118-58: SASC FY24 Committee Report	Page 365	Hydrogen fuel cell research for defense applications	**Appendix L**	**75 – 80**
	Page 358	Benefits and use of electrochromic glass in military bases and installations	**Appendix M**	**81 – 83**
	Page 363	Ground source heat pumps	**Appendix N**	**84 – 87**

Appendix D - Energy Performance Master Plan

D.1. Energy Performance Master Plans

Title 10 U.S.C § 2911(d)(1) and (2) requires the Secretary of Defense and each military department and defense agency, or DoD Component, to develop comprehensive master plans for the achievement of DoD energy performance goals. This has been accomplished through the development of comprehensive installation and energy master plans by each Component.

Installation Energy Plans (IEPs) incorporate installation, Component, and higher-level guidance into a holistic roadmap that enables the installation to work toward its energy efficiency, renewable energy, energy resilience, and climate resilience goals. Energy resilience exercise outputs are combined with analyses from the Defense Climate Assessment Tool and data from the DoD Regional Sea Level Database to inform the IEPs. The IEPs, in turn, are used to inform the installation resilience plans (IRPs), required by title 10 U.S.C. § 2864 (a)(2), which then are incorporated into the installation master planning process and feed into the installation master plan.

Overall, an IEP directs a structured and effective approach to selecting, prioritizing, sequencing, and implementing, energy projects and programs that ultimately result in better long-term installation energy performance. IEPs should be developed in coordination with other installation plans. Every effort should be made to integrate the IEP development process for developing master plans. While the IEP must be compatible with the installation-level master plan, the IEP should be either a stand-alone document or a severable product within the master plan. IEPs should be continuously maintained, incorporating projects' lessons learned and synchronizing with the annual energy reporting process.

D.2. Energy Performance Goals

The DoD energy goals in Tables C-1 and C-2 are set forth by title 42 U.S.C. § 15852(a) and title 10 U.S.C. § 2911(g). These goals focus on renewable energy use. The Department remains committed to maximizing the efficient use of energy to free up resources for higher priorities. With respect to renewable energy, the DoD strives to optimize the use of on-site distributed energy resources from all sources of energy generation to directly improve mission assurance. The type of generation source is determined by local availability, market conditions, a business case, or mission requirements. As such, the Department is committed to optimizing the effective and efficient use of generating sources.

Section 2911(c)(3) of title 10 U.S.C. requires DoD to include installation energy security and resilience goals in this report and subsequent AEPRRs. In FY 2022, DoD components continued use of BSEs and other energy assessments to conduct full-scale and routine testing on the infrastructure and equipment supporting all critical energy requirements and determine energy availability at the mission level. In FY 2023, the Services successfully conducted 19 independent, full-scale exercises. Planning is in progress for additional BSE frameworks, such as cybersecurity modules.

Table D-1. DoD Renewable Energy Goal

Goal	Description	Uniform Measure	Method of Measurement	Metric
Consume More Electric Energy from Renewable Sources Title 42 U.S.C. § 15852(a)	Increase consumption of renewable energy	Installation renewable energy consumption	Total renewable electricity consumption as a percentage of total facility electricity consumption	MWh
Produce or Procure More Energy from Renewable Sources Title 10 U.S.C. § 2911(g)	Increase production or procurement of renewable energy	Electric and non-electric renewable energy production and procurement	Electric and non-electric renewable energy produced or procured compared to total facility electricity consumption	MWh

Table D-2. Renewable Energy Performance Targets

Target	FY11	FY12	FY13	FY14	FY15	FY16	FY17	FY18	FY19	FY20	FY25
Consume More Renewable Energy	+5%	+5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%	+7.5%
Produce/Procure More Renewable Energy [1]	-	-	-	-	-	-	-	+15%	-	-	+25%
[1]FY18 interim target required by Title 10 U.S.C. § 2911(g)(2).											

Table D-3. Critical Energy Availability

Target	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27	FY28	FY29	FY30
100% of critical mission energy load has a minimum of at least 99.9% or higher availability by FY30 10 U.S.C. § 2920	-	-	-*	-	-	-	-	-	-	-	99.9+%
*Specific availability standards between 99.9% and 99.9999% availability per fiscal year (depending on criticality of mission), and interim goals prior to FY25, may be established by the Secretary of Defense pursuant to Title 10 U.S.C. § 2920.											

Table D-4. Black Start Exercises

Target	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27
Each military department will conduct not fewer than 5 BSEs per year through FY27 Title 10 U.S.C. § 2920	-	-	15	15	15	15	15	15

Appendix E - DoD Energy Performance Summary

Table E-1. Renewable Electric Energy

Renewable Electric Energy Requirement per Title 42 U.S.C. 15852(a)	Renewable Electricity Use (MWh)	Total Electricity Use (MWh)	Percentage of Facility Electric Use	EPAct 2005 Requirement
Eligible renewable electricity use as a percentage of total electricity use	2,175,921.0	28,901,230.9	7.5%	7.5%

Table E-2. Procuring Renewable Energy Sources

Produce or Procure More Energy from Renewable Sources per Title 10 U.S.C. 2911(g)	Renewable Energy Produced/Procured (MWh)	Total Electricity Use (MWh)	Percentage of Facility Electric Use	Compliance Target by 2025
Total renewable energy (electric and non-electric) produced or procured as a percentage of total facility electricity consumption	5,177,422	28,901,230.9	17.9%	25.0%

Table E-3. Goals for Buildings Metered for Electricity

Meter Goals	Cumulative # of Buildings Metered for Electricity	Cumulative % of Appropriate Buildings Metered for Electricity	Cumulative # of Buildings Metered for Natural Gas	Cumulative % of Appropriate Buildings Metered for Natural Gas	Cumulative # of Buildings Metered for Steam	Cumulative % of Appropriate Buildings Metered for Steam
Standard Meters in FY23	49,791	44.6%	24,879	38.4%	2,173	6.0%
Advanced Meters in FY23	26,325	23.6%	8,488	13.1%	2,462	6.8%
Total Meters in FY23	76,116	68.2%	33,367	51.6%	4,635	12.8%

Table E-1. Energy Efficiency Standards for Federal Buildings

Federal Building Energy Efficiency Standards	Percentage of New Building Designs	Compliance Target
Percentage of new building designs started since beginning in FY 2007 that are 30% more energy efficient than relevant code, where lifecycle cost-effective (including 8/2012 standards)	76.5%	100.0%

E.1. Investments in Energy Management

Table E-2. Investment Sources for Energy

Sources of Investment	Investment Value (Thou. $)	Anticipated Annual Savings (MMBtu)
Direct obligations for facility energy efficiency improvements	$28,993.0	5,055.2
Investment value of ESPC Task/Delivery Orders awarded in fiscal year	$74,365.3	67,901.0
Investment value of UESC Task/Delivery Orders awarded in fiscal year	$192,563.5	574,533.0
TOTAL	**$295,921.8**	**647,489.2**

Table E-3. Funding for ESPC/UESC

Investment	Percentage
Total Investment as a percentage of total facility energy cost	6.3%
Financed (ESPC/UESC) investment as a percentage of total facility energy costs	5.7%

Table E-4. Installation Energy Type and Cost

Total Installation Energy Consumption and Cost[16]			
Energy Type	BBtu	Cost (thou.)	% of Consumption
Electricity	95,157.45	$3,167,345.78	48.1%
Fuel Oil	13,731.78	$293,101.57	6.9%
Natural Gas	74,725.81	$760,238.74	37.8%
Liquified Petroleum Gas	904.74	$18,292.75	0.5%
Coal	3,957.48	$24,237.73	2.0%
Steam	2,938.34	$117,187.34	1.5%
Other	871.29	$22,882.23	0.4%
Renewable Electric, On-site	3,089.69	$89,681.96	1.6%
Renewable Electric Off-Site	987.93	$5,510.32	0.5%
Renewable, Other, On-Site	1,178.28	$2,969.53	0.6%
Renewable, Off-Site Green Energy Purchases	232.62	$16,906.99	0.1%
Renewable Energy On-site, Replacement REC Adjustment	763.71	$0.00	0.4%
TOTAL BUILDING	**197,775.4**	**$4,518.354.93**	**100%**
Non-Tactical Vehicles	7,107.34	$213,841.88	
TOTAL Building + NTVs	**204,882.75**	**$4,732,196.81**	

[16] Total energy consumption and cost does not include on-site produced, renewable energy replacement REC adjustment.

Appendix F - Recommended Change in Organization or Authority

The Department has no recommendations for changes in organization or authority.

Appendix G - Installation Energy Program Management

G.1. Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization (ODASD(E&RO))

The Assistant Secretary of Defense for Energy, Installations, and Environment (ASD (EI&E)) is within the Office of the Under Secretary of Defense for Acquisition and Sustainment. The Office of the Deputy Assistant Secretary of Defense for Energy Resilience and Optimization (ODASD(E&RO) is within ASD (EI&E) and prescribes policies and procedures, provides guidance, and monitors and reviews programs related to operational energy (OE), installation energy (IE), climate resilience, innovation, and the siting clearinghouse (see Figure G-1).

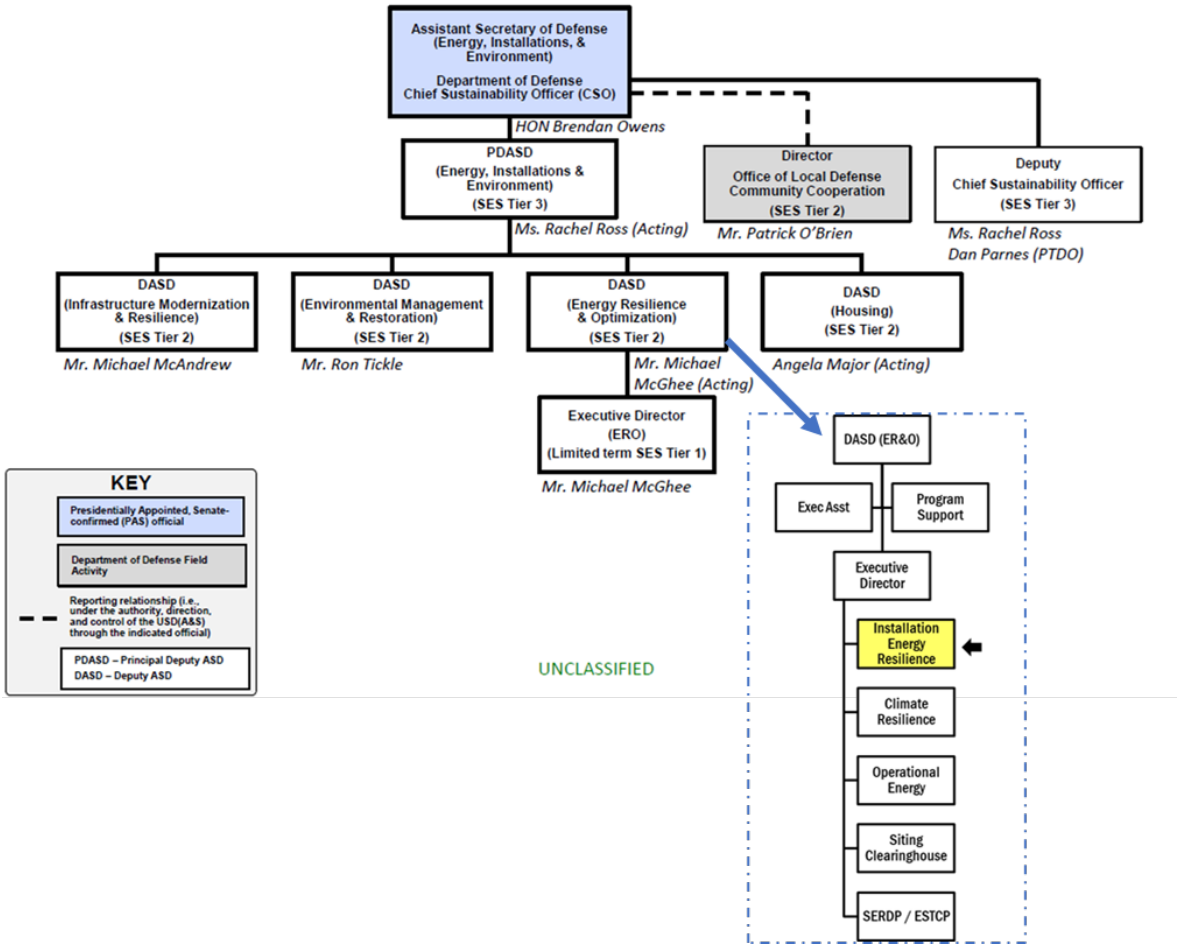


Figure G-1. Department of Defense Energy Management Organizational Structure

The mission of ODASD(ER&O) is to sustain warfighting readiness and lethality by providing all energy-related policy and governance for programs and activities that enable resilient, efficient, and cyber-secure energy for Joint forces, weapon systems, and installations. To accomplish this mission, ODASD(ER&O) supports initiatives across three primary areas:

- **Energy Resilience:** Enhancing the military capability, readiness, and resilience of DoD installations and forces through assured access to resilient and cyber-secure fuel and power;

- **Energy Risk:** Identifying, assessing, and integrating energy-related analyses and risks into Department decision-making associated with requirements, deliberate planning, wargames and exercises, installation master planning, Energy Resilience and Conservation Investment Program (ERCIP), and investments in forces and installations; and

- **Energy Performance:** Ensuring energy efficiency and lower costs at DoD installations through reliable, efficient use of power and alternative financing mechanisms.

G.2. Department of the Army—Assistant Secretary of the Army for Installations, Energy, and Environment (ASA (IE&E))

The Assistant Secretary of the Army for Installations, Energy and Environment (ASA(IE&E)) oversees the Army's energy, water, and sustainability programs, see Figure G-2. The ASA(IE&E) establishes policy, oversees programs, and sets the strategic direction for installations, housing, and partnerships; energy and sustainability; environment, safety, and occupational health; and installation modernization. The Army's senior energy official, the Deputy Assistant Secretary of the Army for Energy and Sustainability (DASA(E&S)), is responsible for overall program direction, establishing policies, developing and refining strategies, and oversight of all programs and initiatives related to energy and water resilience and sustainability within the Army. The Deputy Chief of Staff (DCS), G-9, is the Army's lead integrator for installation matters, including installation readiness and operations, military facilities investment requirements, installation services, privatization and partnerships, energy and water resilience, and sustainability. The G-9 provides installation, energy, and environmental program policy implementation guidance to Landholding Organizations: Army Materiel Command (AMC), Army Central (ARCENT), Army National Guard (ARNG), and the U.S. Army Reserve (USAR). Each Army Landholding Organization implements Army policy and guidance in support of the Army goals and strategy.

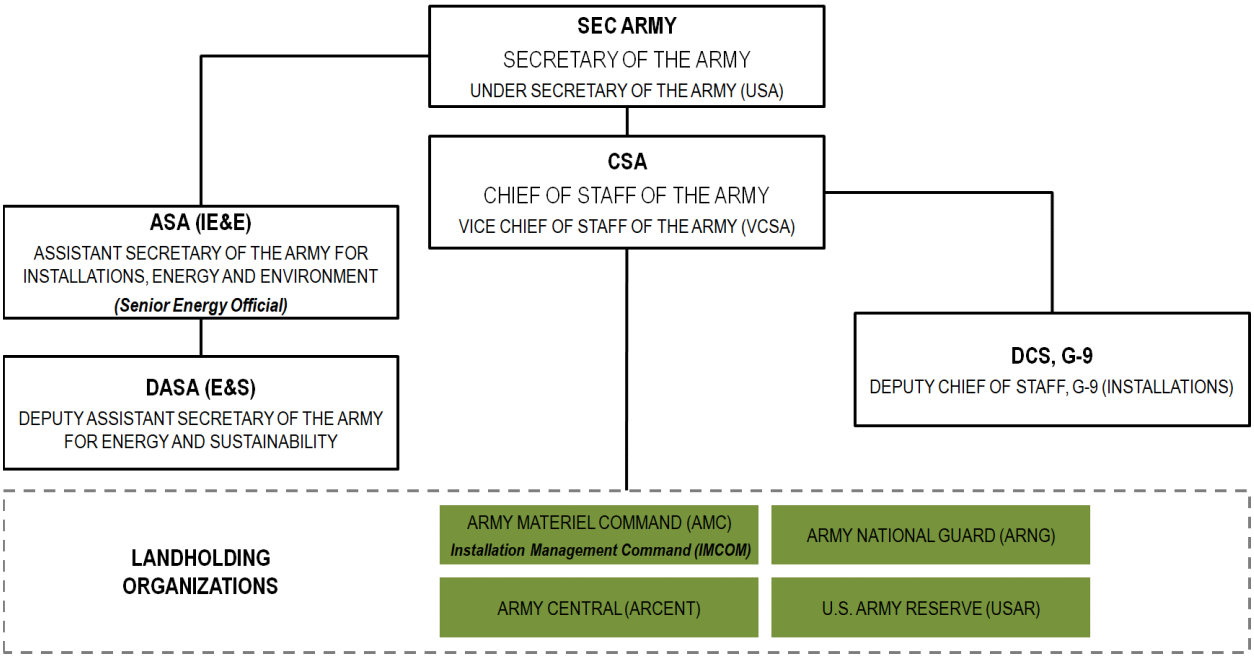


Figure G-2. Department of Army Energy Management Organizational Structure

G.3. Department of the Navy (DoN)—Assistant Secretary of the Navy for Energy, Installations, and Environment (ASN(EI&E))

The Assistant Secretary of the Navy for Energy, Installations and Environment (ASN(EI&E)) serves the Department of Navy (DoN) by enhancing combat capabilities for the warfighter through greater energy security; the acquisition and disposal of real property; construction and maintenance of installations; environmental protection, planning and restoration ashore and afloat; and conservation of natural and cultural resources. The Deputy Assistant Secretary of the Navy for Energy (DASN(E)) is the principal energy advisor to ASN(EI&E). Within the Secretariat, the Director for Installation Resilience is responsible for military installation readiness through mission assurance and resilience. The DoN's program for energy security promotes innovation and optimizes the use of congressional authorities to deliver energy and water resilience on and off DoN military installations. The Navy Shore Energy Management Enterprise Organizational Structure is shown in Figure G-3 below.

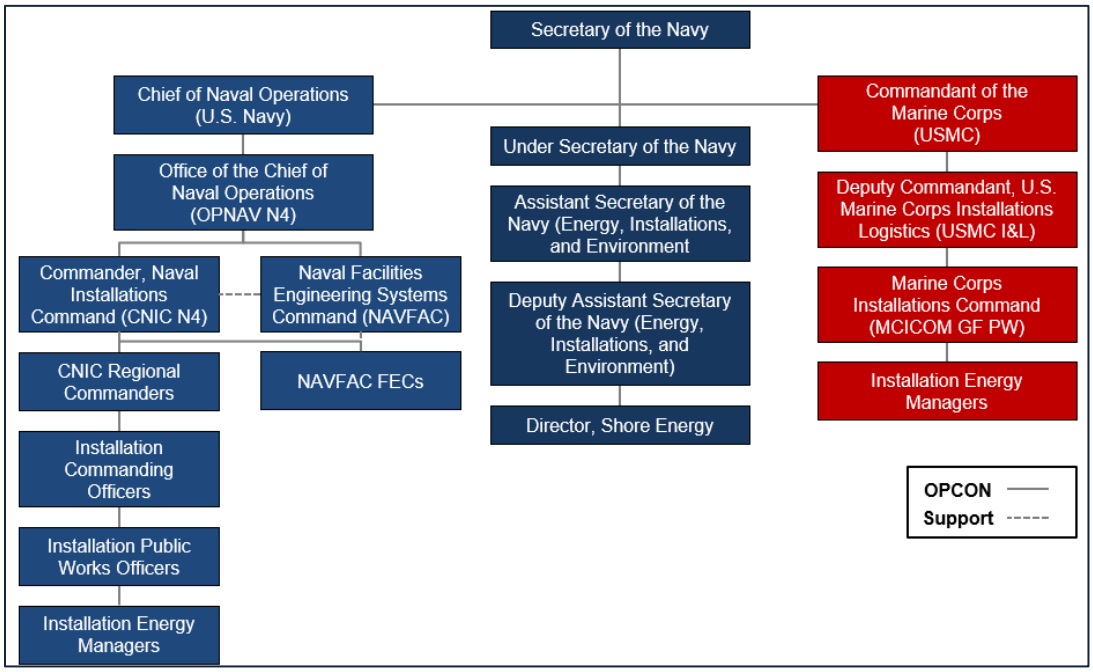


Figure G-3. Department of Navy Energy Management Organizational Structure

The Office of the Chief of Naval Operations (CNO) Installations Division (OPNAV N4I) is responsible for developing policy and programming resources for the DoN's Facility Energy and Mission Assurance Programs. OPNAV N4I also ensures compliance with DoN Shore Energy Goals through OPNAV Instruction 4100.5E (Shore Energy Management). Commander, Navy Installations Command (CNIC) is responsible for leading, managing and integrating the Navy's Budget, Policy, and Requirements across the Navy Shore Energy Enterprise to include current and future shore energy, water, cybersecurity, and mission requirements across warfighter enterprises. The DoN's energy program aligns under CNIC N4 (Facilities and Environmental Department) and N42 (Energy and Base Operations Support (BOS) Division), with the Energy Headquarters Program Director (Energy HPD) responsible for leading and managing the Shore Energy Program across all 10 Regions and 70 Installations. This includes providing budgetary, policy, personnel, and management oversight of diverse programs, projects, and initiatives that directly enhance the Navy's Shore Energy Program.

The Deputy Commandant for Installations and Logistics (DC I&L) is responsible for establishing energy and water management policy for Marine Corps installations in accordance with the Commandant's direction. The Assistant Deputy Commandant for Installations and Logistics (Facilities) serves as the single point of contact responsible for program management and resourcing. The Commander, Marine Corps Installations Command (MCICOM) oversees program planning and execution with direct support provided by the Director, Facilities (MCICOM GF). The Public Works Section (MCICOM GF- PW) includes program management for the Installation Energy, Utilities, Facilities Maintenance, Real Property Services, and Facility-Related Control Systems (FRCS) programs for the Marine Corps.

G.4. Department of the Air Force—Assistant Secretary of the Air Force for Energy, Installations, and Environment (SAF/IE)

Headquarters, United States Air Force Deputy Chief of Staff for Logistics, Engineering, and Force Protection (AF/A4) provides information required to support governance and oversight of energy management activities across DAF installations. The AF/A4 develops guidance, procedures, and practices to enhance DAF energy assurance with the goal of energy resilience and ensures energy security to meet mission-essential requirements. The DAF's organizational structure for energy oversight is shown in Figure G-4.1.

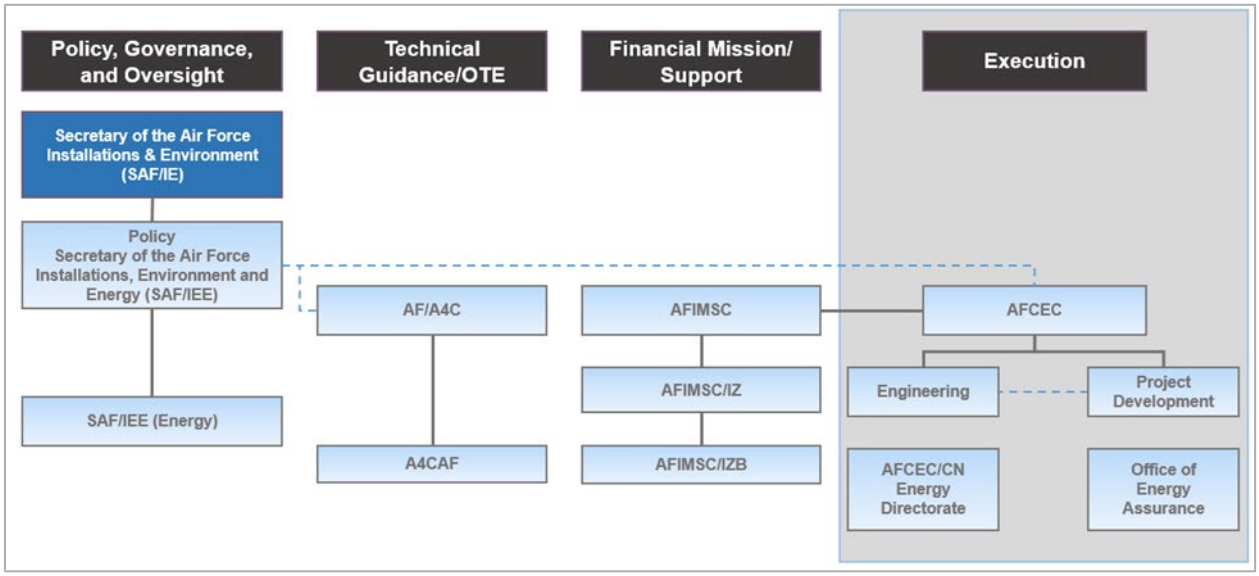


Figure G-4.1. Department of Air Force Energy Management Organizational Structure

Policy: DAF installation energy management is spelled out in Department of the Air Force Instruction (DAFI) 90-1701, Installation Energy and Water Management, last updated in December 2020. DAFI 90-1701 provides guidance on energy and water resource management and resilience, as well as project development and funding. The instruction outlines the specific roles and responsibilities for all individuals and organizations involved in the management of facilities energy and water.

The publication of EO 14057 in December 2021 has led DAF to consider how to incorporate multiple elements of sustainability into its mission-focused approach to installation energy management. As part of the approach to address the targets specified in the EO, DAF published the Climate Action Plan (CAP) in October 2022. The DAF CAP spells out three climate priorities for the Department: 1) Maintain air

and space dominance in the face of climate risks, 2) Make climate-informed decisions, and 3) Optimize energy use and pursue alternative energy sources. In July 2023, DAF published the Climate Campaign Plan, which outlines specific objectives and key results to address the DAF CAP priorities.

DAF issued a memo on Advancing Utilities Privatization Investments in August 2023, and a memo on Achieving Carbon Pollution-Free Electricity (CFE) Targets in November 2023. The Deputy Assistant Secretary of the Air Force Environment, Safety and Infrastructure (SAF/IEE) will be issuing several more memos related to the implementation of EO 14057 in late 2023 or early 2024.

Governance: The DAF Energy Program governance structure provides strategic guidance for development, review, validation, and approval for near- and long-term energy project direction.

Facility Energy Board (FEB): The FEB is the strategic forum used to ensure that the Secretariat and Air Staff vision and priorities are fully incorporated into the DAF Facility Energy Program's decisions and processes. This body ensures that Headquarters Air Force (HAF), DAF Facility Energy Program, and Major Command strategic goals and priorities are considered when balancing risk to mission and risk to Airmen and Guardians within available resources, as well as whether the DAF Corporate Structure should increase energy resiliency efforts. Where practical, the FEB also serves as a forum to discuss industry initiatives and/or best practices in achieving DAF energy goals.

Facility Energy Group (FEG): The FEG serves as the mid-level decision forum to ensure appropriate oversight of facility energy projects. This governance board is focused on ranking projects based on HAF priorities, identifying risks and mitigation plans to be briefed at the FEB, and discussing opportunities for collaboration across organizations to increase efficiencies for project transition and execution.

Facility Energy Panel (FEP): The FEP serves as an action officer-level advisory committee for all DAF energy and water resilience projects. Members of the FEP determine if energy, utility, or water resilience requirements will be transitioned to their appropriate execution partner. The FEP is also responsible for validating the prioritization of the resilient energy and water project portfolio.

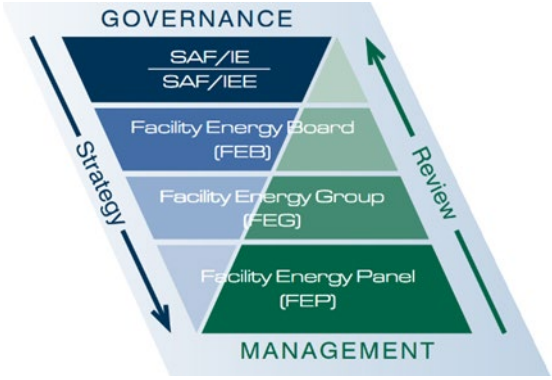


Figure G-4.2. Department of Air Force Energy Management Governance/Management Structure

G.5. Defense Agencies

The Defense Agencies continue to enhance their installation energy management programs and coordinate with stakeholders to address energy and water resilience issues. Where agencies exist as tenants on installations, they work to participate in host-installation energy conservation and resilience

programs. Each has a designated senior energy official to administer their respective program (Table G-1).

Table G-1. Defense Agencies Senior Energy Officials

DoD Component	Senior Energy Official
Defense Contract Management Agency (DCMA)	Energy Program Manager
Defense Commissary Agency (DeCA)	Energy Program Manager
Defense Finance and Accounting Service (DFAS)	Director, Enterprise Management Service
Defense Intelligence Agency (DIA)	Director, Office of Facilities and Services
Defense Logistics Agency (DLA)	Installation Management Director
Missile Defense Agency (MDA)	Environmental Executive
National Geospatial-Intelligence Agency (NGA)	Director, Installation Operations Office
National Reconnaissance Office (NRO)	Chief, Facilities Support Group
National Security Agency (NSA)	Chief of Installations and Logistics
Washington Headquarters Services (WHS)	Pentagon Energy Program Manager